

Multivariable Calculus (MT1008)

Sessional-II Exam

Course Instructor(s):

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Section(s): BCS All Sections

Total Time (Hrs): 1

Total Marks: 70

Total Questions: 6

Date: Apr 10, 2026

Roll No

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Course Section

Student Signature

Attempt all the questions.

[CLO 1: Apply multivariable calculus concepts to solve standard mathematical problems involving partial derivatives, multiple integrals, and basic vector fields.] [10 marks]

Question 1

A robotics team is developing a directional sensor whose detection range varies with angle due to hardware constraints. The sensor's effective range (in meters) is modeled in polar coordinates by

$$r(\theta) = 12 \cos(3\theta); r \geq 0.$$

Due to interference patterns, the sensor produces three identical directional lobes, and the system processes only one lobe at a time. You are required to:

- Calculate the interval of θ corresponding to a single lobe.
- Illustrate a double integral in polar coordinates to represent the coverage area of one lobe.
- Solve the integral to compute the effective coverage area.

[CLO 2: Analyze and solve advanced mathematical problems by integrating multiple techniques of multivariable calculus, such as optimization, and line/surface integrals.] [10 marks]

Question 2

A distributed service runs two types of tasks on a server: CPU-bound (x units) and I/O-bound (y units). The performance cost is modeled by $f(x, y) = x^2 + 4y^2 - 4x + 8y$. Due to a service-level agreement, the weighted throughput must be constant: $x + 2y = 3$.

- Use Lagrange multiplier to calculate constrained minimizer (x^*, y^*) .
- Compute the multiplier λ , and calculate the minimal cost $f(x^*, y^*)$.
- Verify and analyze whether the solution is global minimum.

[15 marks]

Question 3

A machine learning system tunes two parameters x and y to maximize performance score:

$$P(x, y) = 4x - x^2 - y^2,$$

where the parameters are restricted due to resource limits: $x^2 + y^2 = 5$.

- Calculate the critical point(s) of $P(x, y)$ without using method of Lagrange multipliers.
- Calculate where the absolute maximum occurs on the given closed bounded domain.

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Question 4 ✓

[15 marks]

You are tasked with placing a sensor at a coordinate $\bar{x} = [x_1, x_2]^T$ on a circular antenna platform. The goal is to minimize the squared distance (localization error) between the sensor and a fixed target. The optimization problem is defined as:

$$\text{minimize } f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 2)^2$$

$$\text{subject to } g(x_1, x_2) = x_1^2 + x_2^2 - 1 \leq 0.$$

- Outline the Karush-Kuhn-Tucker conditions for this system. Include the Lagrangian function, stationarity, primal and dual feasibility, and complementary slackness.
- Solve the system to **calculate** the optimal sensor location (x_1^*, x_2^*) and the Lagrange multiplier λ .
- Sketch the feasible region and plot both the target point (1,2) and optimal solution (x_1^*, x_2^*) on the same coordinate plane.

[CLO 3: Evaluate application-oriented problems by applying multivariable calculus techniques such as gradient descent and computation graphs in the relevant physical or computational contexts]

Question 5

[10 marks]

You are designing a predictive model for CPU load in a cloud computing environment. The goal is to optimize dynamic resource allocation for multiple servers based on system metrics. The system provides two real-time features for each server:

- x_1 : Number of active user requests per second
- x_2 : Average memory usage (GB)

The predicted CPU load $\hat{y}$ is modeled as a linear function of the features $\hat{y} = w_1 x_1 + w_2 x_2$ and the loss for a single observation is defined as squared error: $L = (\hat{y} - y)^2$ where y is the actual CPU load (%). Using the given values $x_1 = 50, x_2 = 8, w_1 = 0.5, w_2 = 2, y = 50$, construct an artificial neural network (computational graph) and compute the predicted CPU load $\hat{y}$ and loss L by performing a forward pass. Also, derive the gradients $\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}$ using automatic differentiation (back propagation), and **explain** the meaning of each gradient in the context of resource management and system optimization.

Question 6 ✓

[10 marks]

A cybersecurity research team is developing an AI-based intrusion detection system to monitor suspicious activity in a corporate network. The system evaluates two parameters: x (anomaly score based on login behavior), and y (anomaly score based on network traffic patterns). The total threat detection score is modeled by $T(x, y) = x^2 + y^2 - 4x - 6y + 12$. The system starts with parameters (0,0). To improve detection, the engineers apply gradient descent algorithms. **Predict** the updated values of x and y by performing one iteration of gradient descent.